JAVIER CABRERA

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Senior Engineer

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WHO AM I?

Senior Software Engineer with a PhD in Automated Testing, specializing in WebAssembly runtimes and compiler tooling. Today, I build production-grade, low-latency distributed services in Go and Rust, combining deep systems knowledge with large-scale cloud infrastructure experience. My work spans high-throughput voice platforms running in Kubernetes (GCP).

CORE EXPERTISE

- **Languages**: Go, Rust, Python, Java, JavaScript, C/C++
- **Cloud & Infra**: Kubernetes, GCP, Azure, AWS, OVH
- **Data & Streaming**: Kafka, RocksDB, MongoDB, ETCD, Apache Airflow, SQL, RonDB, Clickhouse
- **AI & Runtime**: Argo Workflows, ONNX Runtime, vLLM, Wasmtime
- **Observability**: Prometheus, Grafana, Datadog, StatsD protocol, KEDA
- **CI/CD & DevOps**: Gitlab, GitHub Actions, Firecracker

SELECTED ACHIEVEMENTS

- Identified 5 security vulnerabilities (Filecoin bug hunting program)
- Credited in CVE-2021-32629 (Fastly Inc)
- PhD in Automated Testing (KTH Royal Institute of Technology)

EXPERIENCE

3/2025 – Present — **Senior Software Engineer** — **Sinch AB.**

I design and operate high-throughput, low-latency, real-time voice systems powering large-scale conversational AI platforms. I design and operate production-grade audio streaming services built on WebRTC and Kubernetes (deployed in GCP), supporting thousands of calls per day with hundreds of concurrent sessions. My work includes real-time bidirectional audio streaming, AI-driven voice bots, and cloud-native architectures. I have hands-on experience with Kubernetes operations, custom autoscalers, KEDA, and observability stacks including Datadog, Prometheus, and Grafana. I also led the production deployment of a cost-efficient Voice Activity Detection system based on Silero v5 models, achieving an average inference latency of 6ms for 32ms audio chunks at scale.

1/2024 – 3/2025 — **Software Engineer** — **Hopswoks AB.**

I led the migration from on-premise deployments to a generic Kubernetes-based solution, enabling fully installable, production-ready Hopsworks clusters across GCP, AWS, Azure, and OVH. I contributed extensively to DevOps and MLOps initiatives, designing and implementing custom automated test frameworks, including fully simulated air-gapped environments to validate end-to-end Kubernetes installations. My work spanned CI/CD pipelines, cloud-native infrastructure, and distributed systems, with experience in Kafka, Arrow Flight, RonDB, and large-scale MLOps platform operations.

3/2019 – 3/2024 **Phd/Researcher/Teacher** **KTH Royal Institute of Technology.**

I was a PhD candidate, researcher, and teaching assistant at KTH Royal Institute of Technology, where I earned my doctorate in Software Diversification with a focus on WebAssembly, automated testing, and software engineering. My doctoral research advanced methods for automatic software diversification and binary transformation to improve the reliability, security, and resilience of WebAssembly applications, producing tools and techniques such as Wasm-Mutate, and publishing multiple peer-reviewed articles in top venues. I also contributed to edge computing and compiler tooling, taught undergraduate courses, and worked extensively with automated testing.

11/2022 – 4/2023 Concurrent with PhD research **Contractor Automatic Testing Engineer** **Filecoin Foundation.**

I worked as a senior automated testing consultant for the Filecoin Virtual Machine (FVM), the WebAssembly-based runtime powering smart contracts on the Filecoin blockchain. I designed and implemented advanced testing and fuzzing strategies targeting the Wasmtime compiler used by the FVM, strengthening the reliability and security of smart contract execution. During this engagement, I was credited with identifying five security vulnerabilities. I also pioneered (in their stack) the use of autoencoder-based anomaly detection techniques to uncover subtle compiler inconsistencies and unexpected execution behaviors.

09/2021 – 12/2021 Concurrent with PhD research **Contractor Software Engineer** **Fastly Inc.**

I worked on WebAssembly compiler tooling, automated testing, and security fuzzing for Compute@Edge. During this engagement, I contributed to identifying a critical compiler bug in the Cranelift backend used to compile WebAssembly modules for Fastly's edge platform, which was documented by the company and credited in a Fastly security advisory related to CVE-2021-32629. My work involved Rust, libFuzzer, and systematic automatic testing strategies to exercise and harden compiler invariants, helping ensure the reliability and memory safety of Wasm execution at the edge before the issue could affect customers.

11/2017 – 3/2019 **Software Engineer/Full stack developer** **Iberant SL.**

I designed and developed cloud-based applications using .NET Core and a microservices architecture. I built and maintained backend services integrated with MS SQL Server, while also developing responsive frontend interfaces with ReactJS and cross-platform mobile applications using Xamarin. My work included deploying and operating services in Microsoft Azure, contributing to scalable, production-ready solutions across web and mobile platforms.